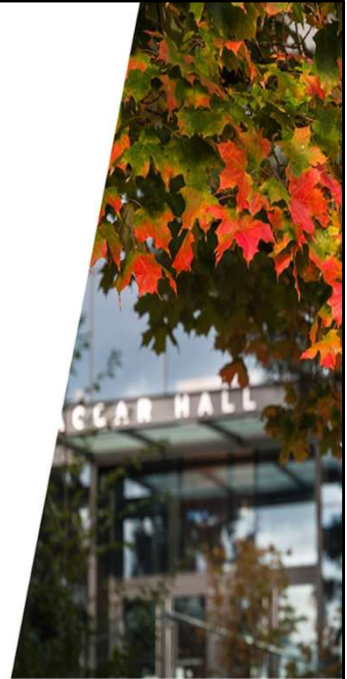




ENRON FORENSIC EMAIL ANALYSIS, CONCLUSION, AND DELIVERABLE

Data Analytics for Professional Accountants
(ACCTG 522)

Class 6 | MPAcc Class of 2026



Welcome!

Class Agenda

- **Case Review: Enron Emails Case**
 - RegEx and DateTime parsing functions
 - Sentiment Analysis and Phrase Analysis
- Case Deliverable Reminder
- Case Workflow Overview:
 - Modifying the Workflow (Capture Individuals of Interest)
- Lab Forensic Analysis & Deliverable write-up

Case Review: Enron Emails Case

Parsing Text using Regex

Our test email:

Message-ID: <25828831.1075855376669.JavaMail.evans@thyme> Date: Mon, 26 Nov 2001 07:25:16 -0800 (PST)
From: k..allen@enron.com To: jsmith@austintx.com Subject: RE: Additional properties in San Antonio Jeff, Can you
resend the info on the three properties you mailed and the one you faxed on Tuesday. I was out of the office last
week. Phillip -----Original Message----- From: "JEFF SMITH" <jsmith@austintx.com>@ENRON Sent: Tuesday,
November 20, 2001 3:43 PM To: Allen, Phillip K. Subject: Additional properties in San Antonio Phillip, I am waiting
to get info. on two more properties in San Antonio. The broker will be faxing info. on Monday. One is 74 units for
\$1,900,000, and the other is 24 units for \$550,000. Also, I have mailed you info. on three other properties in
addition to the 100 unit property that I faxed this AM. Let me know if you have any questions. Jeff Smith The Smith
Company 9400 Circle Drive Austin, TX 78736 512-394-0908 office 512-394-0913 fax 512-751-9728 mobile
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Unique Key

Unstructured data

Key identifier variables

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What patterns can we use to extract meaning from the Subject/Body of the email (just this email)?

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Indicates it is a Reply

Helps us distinguish the reply from the original email

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Discussions involving (parsable) cash amounts

Various parsable contact details, emails, phone #s

And address

As well as specific keywords like "properties" "broker" and phrases like "I was out of the office last week"

RegEx Sensitivities

Sentiment analysis:

We use `\<\w+\>` (could also use `[[:word:]]+`) requires correct spelling of each word and a match in the dictionary.

We do match 4,179 words but don't match 116,959 "words" (with numbers, missing spaces, typos etc)

RegEx Sensitivities

Sentiment analysis:

Alternative is to try and remove all the numbers and other characters and see if there are actual words of interest, e.g.:

010413_Termination

RegEx Sensitivities

Phrases on unstructured text blocks:

We often use `.*\s([what we want to capture])\s.*`

Which expects zero or more “anything” followed by a space.

My password is:pa55word

versus

My password:pa55word.

Using OR

Another way to make the expressions more robust is to use the or command “|”

`(blue|red|yellow)`

`\.|\\s|\\(`

`Password.xyz Password xyz Password(xyz etc`

Keyword Sensitivities

When using the join approach (like in our case)
Keywords require exact match on wording e.g.,
Keyword = "Fraud" only matches "Fraud" not "Frauds",
"Defraud", "Fraudulence" or "Fraudulent" or lowercase
versions (i.e., it is case sensitive).

Alternative: Use filters and "Contains" formula for fraud
or FRAUD and lowercase/uppercase both fields.

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Memo Structure

Interpretation and Recommendations

Explain significance of the findings, widespread evidence? Identifies very targeted group, discuss why the analysis is important.

Conclude with identification of individuals and suggest further investigative steps **(e.g., deeper audit of certain users as seen in screenshot /table in the attachments).**

Attachments

The Unique Tool

The screenshot displays the Alteryx software interface. The top toolbar includes various tool categories: Favorites, Recommended, In/Out, Preparation (highlighted), Join, Parse, Transform, In-Database, Reporting, Documentation, Spatial, Machine Learning, Text Mining, Computer Vision, Interface, and Data Investigation. The 'Unique' tool icon is highlighted in the bottom right of the toolbar.

On the left, the 'Unique (181) - Configuration' panel is open. It shows the 'Select columns to find unique values from:' section with a list of column names. The 'emailFrom' column is selected, indicated by a green checkmark and a yellow highlight.

On the right, a workflow diagram is shown. It features a 'Unique' tool (represented by a blue icon with a snowflake) connected to other tools. A yellow arrow points from the 'Unique' tool to the 'emailFrom' column in the configuration panel. Another yellow arrow points from the 'Unique' tool to a text box.

Identify the individual email addresses of interest
("U" node outputs a single Unique value for our list; "D" is the duplicates)

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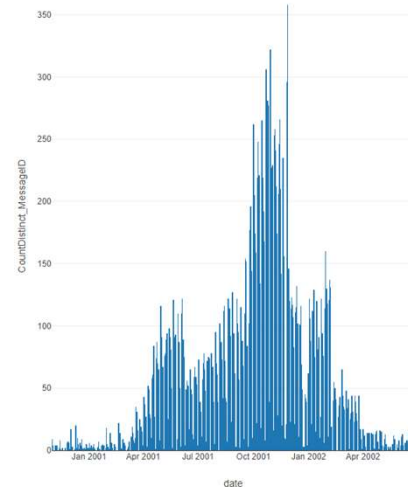
**VERYTHING'S
BIGGER IN**



TEXAS

Enron Key Timeline

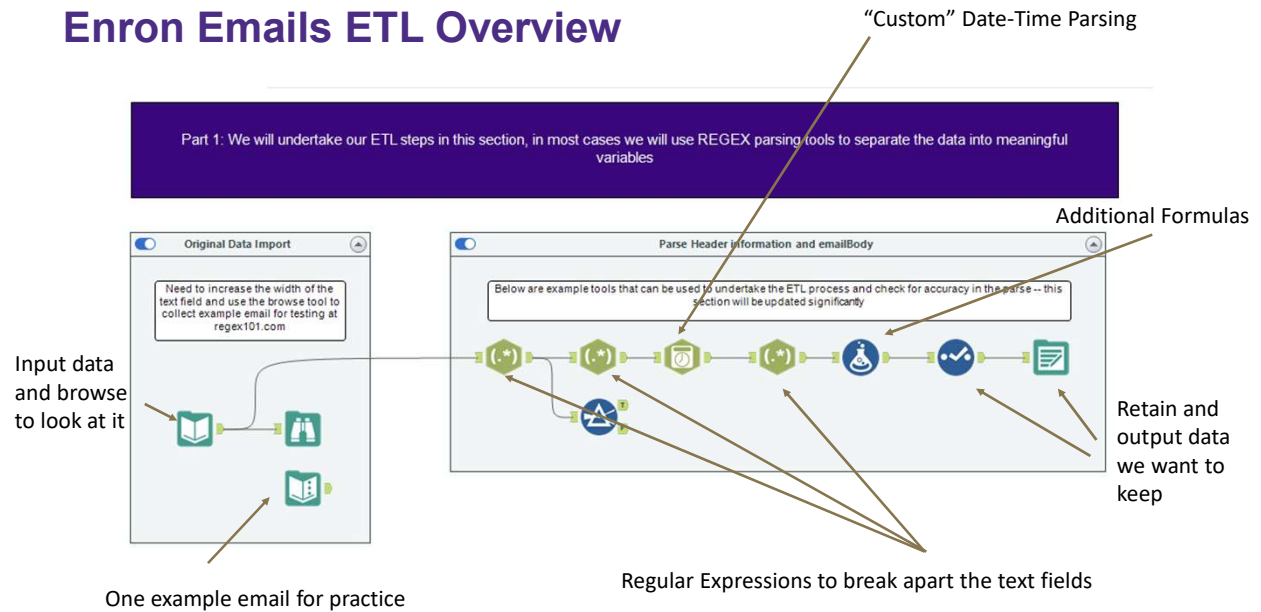
- 1999–2000 → Enron at peak: rapid growth, accounting complexity.
- **Aug 2001** → **Skilling resigns, Watkins memo.**
- **Oct 2001** → **Massive loss and equity reduction disclosed.**
- **Oct–Nov 2001** → **SEC investigation begins.**
- **Nov 2001** → **Restatement and Dynegy merger collapse.**
- **Dec 2001** → **Bankruptcy (at time biggest ever).**
- 2002 → Andersen indicted and collapses.
- 2002–2006 → Enron Task Force prosecutes key executives.



Case Deliverable Question

Conduct a forensic review of internal communications from the Enron email corpus. What evidence, if any, points to failures of governance, accountability, or integrity within the organization? Submit a memo and attach supporting exhibits from your analysis.

Enron Emails ETL Overview



DateTime (25) - Configuration

Select the format to convert

☐ Date/Time format to string

☒ String to Date/Time format

Select the string field to convert

emailDate

Specify the new column name

DateTime_Out

Specify your DateTime Language

English

Select the format that matches the incoming string field

MM/dd/yy h:mm:ss

dd/MM/yyyy h:mm:ss

dd/MM/yy h:mm:ss

day, dd Month, yyyy

dd-MM-yy

dd-MM-yyyy

dd-Mon-yy

dd Month, yyyy

dd-MM/yy

dd/MM/yyyy

dy, Month dd, yyyy

MM-dd-yy

MM-dd-yyyy

MM/dd/yy

MM/dd/yyyy

Month dd, yyyy

Month, yyyy

yyyy-MM-dd

yyyyMMda

yyyy/MM/dd

HH:mm:ss

Custom

Specify the format of the incoming string field

dy, d Mon yyyy HH:mm:ss

Example

Sun, 2 Jan 2000 03:04:05

Output

2000-01-02 03:04:05

Note: The incoming string should match the example.

"Custom" Date-Time Parsing

Specifying: dy, d Mon yyyy HH:mm:ss

**Warning: Sometimes this tool will default back to unconfigured, losing the info in Custom. Type in the date format above exactly to have it run correctly.

Enron Emails Analysis Overview

Part 2: We will undertake our analysis steps in this section, in most cases we will be using either REGEX parsing tools or regex matches to identify the possible cybersecurity related violations.



Save dataset from part 1 and read it into Alteryx

Question 2

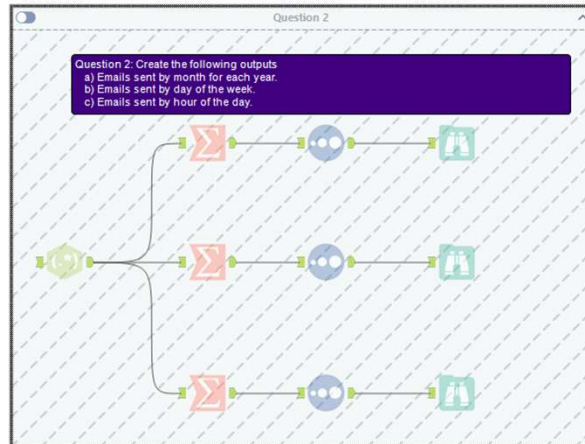
Question 4

Question 3

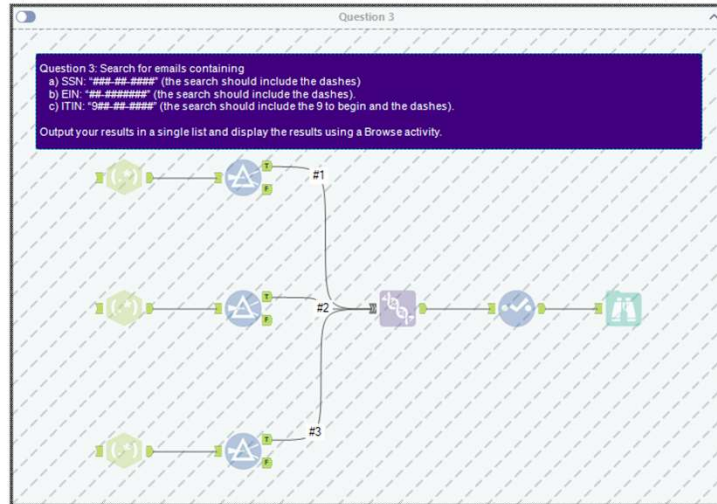
Question 5

Additional Examples

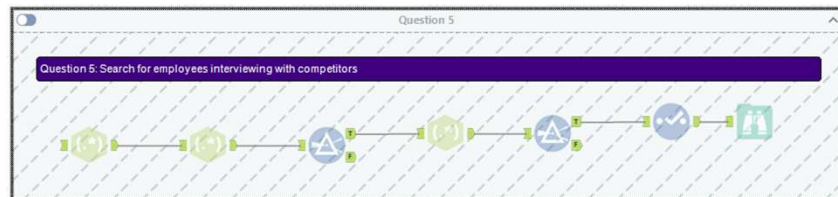
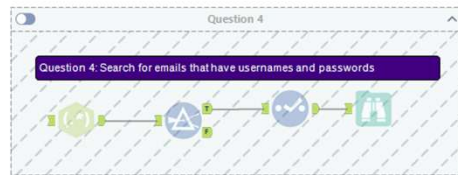
Enron Emails Analysis



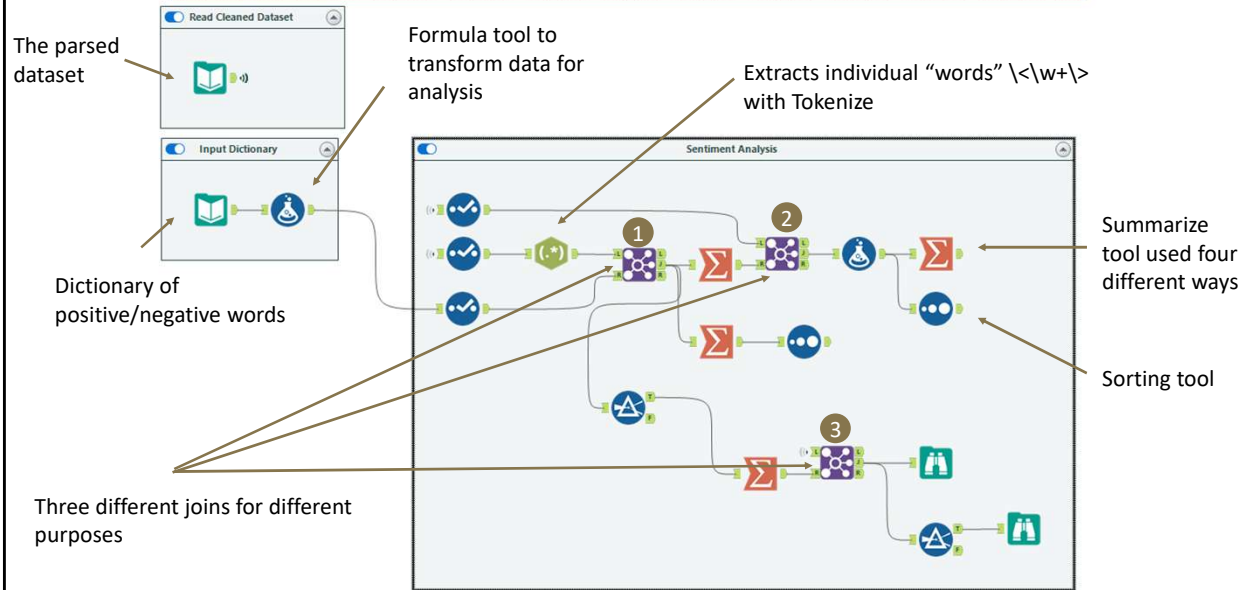
Enron Emails Analysis



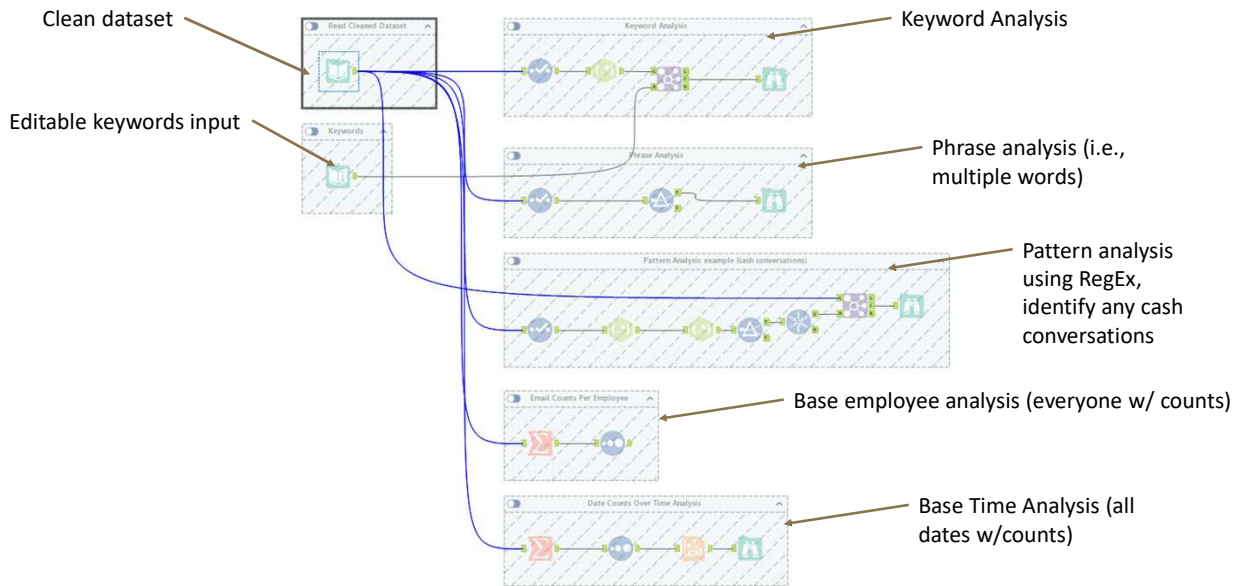
Enron Emails Analysis



Part 3: Examine email tone or sentiment over time and identify key words to search for suspicious or target emails. In this example, we will calculate negative sentiment as negative words / total words, and look for email relating to SEC investigations.



Part 4: Keyword searches and additional tools to help undertake Forensic approach to analysis. find suspicious email conversations, potential ethical concerns or discussions of SEC investigations. As a bonus, see if you can find the poem! Hint: One Fish, Two Fish, Yellow Fish, Enron



Conclusion and look-ahead

We have undertaken a sophisticated analysis of unstructured data.

Please submit your case deliverable by Monday.

Thank you!

